

Read quality in the PDAC KRAS dataset

WEEK 2 PROGRESS REPORT

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Quality-check version: 34 files reviewed and six pilot alignments completed.

Wording revised on 23 September 2026. Results retain the original reporting stage.

Purpose and samples

This stage checks whether the reads are ready for genome alignment. The 17 paired-end runs represent eight cell lines under control or KRAS siRNA treatment, with one extra MiaPaca-2 KRAS run whose repeat type remains unclear.

Methods

Sample identities were checked against the European Nucleotide Archive (ENA). FastQC 0.12.1 reports covered all 34 files: 32 reports from Galaxy and two from complete local recovery files. MultiQC 1.35 combined the reports.

HISAT2 2.2.3 pilots aligned the first 1,000,000 pairs from Control M1 and KRAS P1 to human hg38. Each input was tested with unstranded, forward-stranded (FR) and reverse-stranded (RF) settings.

FastQC results for 34 files

FastQC check	Pass	Warn	Fail	Files
Per-base quality	34	0	0	34
Per-tile quality	34	0	0	34
Per-read quality	34	0	0	34
Base composition	0	0	34	34
GC distribution	32	2	0	34
N content	34	0	0	34
Read length	34	0	0	34
Duplication	0	0	34	34
Overrepresented sequences	34	0	0	34
Adapters	34	0	0	34

Files contained 37,886,040-47,840,900 reads each, all 150 bases long, with 50%-52% GC. Mate counts matched across the 17 runs. MultiQC included 34 entries and showed 57.0%-67.3% duplication. The two GC warnings belong to SRR24828471.

Pilot alignment results

The six pilots each used the first 1,000,000 read pairs from Control M1 or KRAS P1. Overall read alignment was 96.63% for Control M1 and 96.40% for KRAS P1 under all three strand settings. Control M1 had 86.40% concordant unique pairs in unstranded mode and 86.44% in FR and RF modes; KRAS P1 had 87.43% under each setting.

These subsets describe pilot performance, not the later full alignments. Control M1 and KRAS P1 are different cell lines, so their difference cannot isolate a treatment effect. Similar FR and RF rates did not determine strand specificity.

Quality checks and file recovery

Library preparation can affect base composition, while abundant transcripts and amplification can produce repeated sequences. The quality flags do not identify their cause. No trimming or automatic duplicate removal was applied on the basis of these flags. The GC warning alone does not establish contamination.

Incomplete imports were excluded from analysis and retained in the history. Complete local files restored quality-report coverage. SRA Toolkit recovered both Galaxy mates, and an independent FastQC run confirmed their read counts and quality flags.

Work remaining at this stage

The next steps were to establish strand specificity and complete the full alignments. Gene counts and differential expression were not yet available. Later versions of the report contain the full-run results.

References

1. Galaxy FastQC reports, MultiQC 165, HISAT2 pilot summaries 122-136 and local recovery checks, recorded on 22 September 2026. Analysis records: github.com/yichuanlin07/pdac-kas-rnaseq.
2. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.
3. FastQC documentation. bioinformatics.babraham.ac.uk/projects/fastqc/.

Author contribution

Yichuan Lin organized the data and coordinated the analysis and report preparation.

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